Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_175847_m
- started_at_local: 2026-05-13 17:58:47
- finished_at_local: 2026-05-13 18:06:17
- duration_hhmm: 00:07
- duration_seconds: 449.849
- seed_offset: 12000
- replicate_label: m
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.183215	0.041579	0.112397	0.0	0.78	0.0
atsp_random_replay	random	score_only	False	0.18982	0.008454	0.099137	0.795527	0.58	0.013731
atsp_failure_replay	failure	score_only	False	0.181339	0.009004	0.095172	0.853419	0.58	0.014278
atsp_failure_replay_compression	failure	novelty_gate	True	0.193143	0.009004	0.101074	0.813678	0.58	0.018837

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.183215, synthetic 0.041579, combined 0.112397.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.183215 across 4 instances; family means: ft=0.27386, ftv=0.295102, p=0.018149, ry=0.14575.
- Panel synthetic_holdout mean gap 0.041579 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1750, "family": "ftv", "name": "ftv33", "optimality_gap": 0.360809}, {"best_known_cost": 1530, "cost": 2042, "family": "ftv", "name": "ftv38", "optimality_gap": 0.334641}, {"best_known_cost": 1473, "cost": 1798, "family": "ftv", "name": "ftv35", "optimality_gap": 0.220638}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.18982, synthetic 0.008454, combined 0.099137.
- Accepted-epoch count 4, mean accepted code novelty 0.795527, and final complexity 0.58.
- Adaptation efficiency 0.013731 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.18982 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004982, ry=0.097143.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.181339, synthetic 0.009004, combined 0.095172.
- Accepted-epoch count 4, mean accepted code novelty 0.853419, and final complexity 0.58.
- Adaptation efficiency 0.014278 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.181339 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.063237.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193143, synthetic 0.009004, combined 0.101074.
- Accepted-epoch count 3, mean accepted code novelty 0.813678, and final complexity 0.58.

- Adaptation efficiency 0.018837 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193143 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.110456.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Summary of Results for Replay-Aware ATSP Benchmark Suite

Condition	Replay Mode	TSPLIB Gap (Heldout)	Synthetic Gap (Holdout)	Transfer Gap (TSPLIB & Synthetic)	Code Novelty (Mean)	Compression Pressure	Notes on Transfer & Novelty
atsp_no_replay	None	0.183215	0.041579	0.112397	0.0	No	Baseline transfer performance; no code novelty
atsp_random_replay	Random replay	0.18982	**0.008454 (best synthetic)**	0.099137	0.796	No	Improved synthetic transfer; code novelty high but no TSPLIB gap improvement
atsp_failure_replay	Failure replay	**0.181339 (best TSPLIB)**	0.009004	**0.095172 (best transfer)**	0.853	No	Best transfer results; significant code novelty present
atsp_failure_replay_compression	Failure replay + Compression	0.193143	0.009004	0.101074	0.814	Yes	Slightly worse transfer than failure replay without compression; high code novelty

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Key Interpretations

- **Transfer Metrics Priority**:

The main evidence for transfer capability lies in the TSPLIB and synthetic holdout gaps:

- *atsp_failure_replay* provides the lowest TSPLIB mean gap (0.181339) and best combined transfer gap (0.095172), marking it as the best overall for meaningful transfer on external benchmarks.
- *atsp_random_replay* shows superior performance on synthetic holdout gap (0.008454), but TSPLIB transfer gap is higher (0.099137 vs. 0.095172) and TSPLIB gap is worse (0.18982 vs. 0.181339).
- No replay yields worse transfer gaps overall, confirming the benefit of replay.

- **Code Novelty vs. Transfer**:
- Both **atsp_random_replay** and **atsp_failure_replay** conditions yield high code novelty (~0.8+), much larger than no replay (0.0).

- However, improved code novelty does not guarantee improved TSPLIB gaps since random replay exhibits higher novelty but does not outperform failure replay on TSPLIB benchmarks.
- Hence, lexical/code novelty alone does not imply algorithmic invention or better transfer without supporting gap improvements.

- **Replay Modes Distinction**:
- **No replay**: baseline, no memory of prior cases, lowest transfer performance.
- **Random replay**: improves synthetic gap significantly and code novelty but slightly worsens TSPLIB gap vs. failure replay.
- **Failure replay**: prioritized replay of failure cases yields best TSPLIB transfer gaps, better generalization, and high novelty.
- **Failure replay + compression**: compression pressure slightly degrades transfer performance compared to failure replay alone, despite high novelty.

- **Compression Pressure Impact**:
- Compression-aware failure replay reduces mean transfer efficiency marginally (transfer gap 0.101074 vs. 0.095172 without compression) while maintaining code novelty high. This suggests compression pressure may mildly degrade transfer performance.

Conservative Conclusion

- **Failure replay** condition is the most effective for transfer, based on held-out TSPLIB and synthetic gap metrics, supporting the use of failure memory for replay.
- **Random replay** improves synthetic holdout gap drastically and code novelty but offers less benefit or slight worsening for TSPLIB transfer gaps.
- Code novelty increases markedly with replay conditions but does not inherently indicate algorithmic advancement without corroborating improvements in transfer-related metrics.
- Compression-aware replay does not improve transfer metrics beyond failure replay alone and may slightly hinder it.
- Rankings for transfer:

1. atsp_failure_replay
2. atsp_random_replay
3. atsp_failure_replay_compression
4. atsp_no_replay

If prioritizing transfer and optimality gap improvements, **failure replay without compression** should be considered the best strategy among those tested.